

Technical Specifications: USRP B200mini

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1 Gains

The USRP B200mini uses the AD9364 transceiver. Through the UHD driver, the multiple physical gain stages are abstracted into a single programmable gain-amplification parameter (PGA).

- **RX Gain (PGA):** Range from **0.0 dB to 76.0 dB**. Controls the receive chain, including the LNA and baseband stages.
- **TX Gain (PGA):** Range from **0.0 dB to 89.8 dB**. Adjusts the RF signal output power.
- **AGC Modes (RX):**
 - *Manual:* The user defines a fixed gain value.
 - *Fast Attack:* Fast automatic adjustment for burst-type signals.
 - *Slow Attack:* Slow automatic adjustment for constant-power signals.

Operating note: It is recommended to use at least half of the available gain to obtain a reasonable dynamic range.

2 Hardware Filters

The USRP B200mini combines analog baseband filters with digital filters automatically configured by UHD.

1. Analog filters (Baseband LPF)

Modifiable through: `set_rx_bandwidth(bw)` and `set_tx_bandwidth(bw)`.

What it does: Directly adjusts the Butterworth low-pass filter. Any value between 200 kHz and 56 MHz can be set. If it is not configured manually, UHD attempts to adjust it automatically according to the `sample_rate` value.

2. Digital filters (Half-Band and FIR)

Modifiable through: They are not modified directly. They are configured implicitly when using `set_sample_rate(rate)` and `set_master_clock_rate(rate)`.

What it does: UHD automatically calculates the required interpolation or decimation stages and configures the Half-Band filters and digital FIR coefficients to prevent aliasing. Everything happens in the background.

In code, it is enough to set the bandwidth (`bandwidth`) and sample rate (`sample_rate`); UHD performs the required calculations and programs the AD9364 transceiver.

3 Frequency Ranges

Nominal operating range: **70 MHz to 6 GHz**.

4 Sample Rate

Maximum sample rate: **61.44 MS/s** (single-channel SDR).

5 Bits (Resolution)

12-bit ADC and DAC resolution.

6 IQ & DC Corrections

The USRP B200mini already has native I/Q imbalance and DC leakage (*DC offset* or *DC spike*) corrections implemented. The system handles them at different levels:

- **At the hardware level (AD9364 transceiver):** The chip integrates tracking algorithms that continuously correct *DC offset*, both in the RF stages and in baseband, and they are enabled by default. Additionally, the transceiver has integrated I/Q correction through matrix multipliers.
- **At the software level (UHD driver):** The UHD abstraction enables and manages these automatic factory corrections, so it is not necessary to program corrective filters from scratch in user code.
- **Specific calibration:** If automatic adjustment is not sufficient, UHD includes native command-line utilities, such as `uhd_cal_tx_dc_offset` or `uhd_cal_rx_iq_balance`, which allow test signals to be injected and static calibration tables stored in the hardware to fine-tune the values.

In summary, these corrections work *out of the box*, but manual calibrations can be forced through the driver if the application requires it.

7 SDR Inputs and Connectors

The USRP B200mini is a 1x1 full-duplex SDR. Its main connections are concentrated in the USB 3.0 interface, the RF ports, and the external reference input for synchronization.

Connector	Function	Main notes
USB3	Power and data	USB 3.0 interface to the host computer.
J1 TRX	TX/RX RF	Shared TX/RX port; TX output up to +20 dBm.
J2 RX2	RF reception	Dedicated receive input; maximum input power -15 dBm.
J3 Ref/PPS	External reference	Input for 10 MHz reference or PPS pulse; +15 dBm max.
J5 JTAG	FPGA JTAG	Access to the Spartan-6 FPGA for development and diagnostics.
J6 GPIO	Digital control	GPIO pins connected to the FPGA for auxiliary signals.

- **Internal clock:** The master clock feeds the RF and DSP chains. UHD can select it automatically according to the requested sample rate, or the user can fix it through `master_clock_rate`. The typical configurable range is 5 MHz to 61.44 MHz.
- **External Reference Clock:** The J3 input allows the SDR to be disciplined with an external 10 MHz reference. This is useful when frequency coherence between equipment or synchronization with a laboratory source is required.
- **External PPS:** The same J3 connector can be used as a PPS input to align the equipment time reference. In multi-SDR configurations, it is recommended to explicitly fix the `master_clock_rate` to avoid automatic changes that break synchronization.

- **RX port:** J2 RX2 is the dedicated receive input. Applying more than -15 dBm should be avoided to protect the RF front end.
- **TX/RX port:** J1 TRX works as a shared transmit and receive port. During transmission it must be connected to an antenna or 50 Ω load; in loopback tests, using at least 30 dB of attenuation is recommended.

8 Safe Operating Limits

Critical warnings - read before connecting the equipment

- **Maximum RX power:** Never apply more than **-15 dBm** to any RF input (J1 TRX or J2 RX2). Exceeding this level can permanently damage the front end.
- **Loopback:** In TX→RX loopback configuration, use **at least 30 dB of attenuation** between the ports to protect the input.
- **TX port without antenna:** The J1 TRX port must always be terminated with an antenna or a **50 Ω** load when transmitting. Transmitting with the port open can damage the amplifier.
- **External reference J3:** 10 MHz reference signal with a maximum level of **+15 dBm**.
- **ESD and handling:** Always handle with antistatic protection. Do not allow metal objects to touch the PCB while it is powered.
- **Maximum TX power:** The output of the J1 TRX port can reach **+20 dBm** (peak, frequency-dependent); the typical guaranteed value is >10 dBm.

9 Master Clock Rate - Rules and Pitfalls

The `master_clock_rate` (MCR) is the most critical configuration parameter for extracting maximum performance from the device and the main source of silent problems.

Range and valid values

- Configurable range: **5 MHz to 61.44 MHz**.
- Values above 56 MHz are possible but **not recommended**.
- The MCR must be an **integer multiple of the sample rate**. If this condition is not met, UHD displays a warning and internally applies a rate different from the requested one.

Automatic vs. manual behavior

By default, UHD selects the MCR automatically according to the requested `sample_rate`, choosing the highest possible value to enable the largest number of Half-Band filters. Automatic mode is disabled as soon as `set_master_clock_rate()` is called or `master_clock_rate` is passed as an initialization argument.

```
# Set MCR manually (Python UHD)
usrp.set_master_clock_rate(32e6)

# Or through a device argument
uhd_usrp_probe --args="master_clock_rate=52e6"
```

Known pitfalls

- **MCR change breaks synchronization:** Any MCR change reconfigures the entire clock chain, invalidating previous time configurations (PPS, multi-USRP sync).
- **Multi-USRP:** When synchronizing multiple units, **fix the MCR explicitly** on all of them. Do not depend on automatic mode, since it can select different values per unit.
- **Locked MCR prevents sample-rate changes:** If the MCR is fixed and a `sample_rate` that is not an integer submultiple is requested, UHD will apply the nearest valid rate without making it clearly visible.

10 Indicator LEDs

The LEDs allow device status to be diagnosed in the field without needing a console.

LED	Function	States
PWR	Power	Off = no power On = powered (USB or external)
TRX	TX/RX activity	Off = no activity Green = receiving Red = transmitting Orange = alternating TX/RX
RX2	RX2 activity	Off = no activity Green = receiving
S0	Reference lock	Off = no lock Green = reference locked
S1	Reference presence	Off = low or absent signal Green = reference signal present

Interpretation of blinks on TRX/RX2:

- A blink on the TX LED indicates **underflow**: the host is not delivering samples to the SDR on time. It is reported in the console as “U”.
- A blink on the RX LED indicates **overflow**: the host is not consuming samples from the SDR on time. It is reported as “O”.
- If this occurs, reduce the `sample_rate`, increase the system buffer, or review CPU/USB load.

11 GPIO Pinout (J6)

Connector J6 exposes 8 GPIO lines connected directly to the Spartan-6 FPGA, operating at **3.3 V**. It is useful for control signals, external trigger, or integration with auxiliary hardware.

Usage notes:

- GPIO_0 to GPIO_7 lines are configurable as input or output from UHD through the GPIO API of `multi_usrp`.

Pin	Signal	Level	Pin	Signal	Level
1	3.3 V (VCC)	Power	7	3.3 V (VCC)	Power
2	GPIO_0	3.3 V logic	8	GPIO_4	3.3 V logic
3	GPIO_1	3.3 V logic	9	GPIO_5	3.3 V logic
4	GPIO_2	3.3 V logic	10	GPIO_6	3.3 V logic
5	GPIO_3	3.3 V logic	11	GPIO_7	3.3 V logic
6	GND	Reference	12	GND	Reference

- Direction and value control is performed with `set_gpio_attr()` by specifying bank "FP0".
- For applications that require UART signals over GPIO, a custom FPGA bitfile must be compiled.